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LOT SYSTEMS CORPORATION
COSMO® COMPUTER – ON-DEVICE FIRMWARE SPECIFICATION
CLASS      RESTRICTED // S-2 EYES
S-2        VADIK MARMELODOV
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DOCUMENT    LOT-COSMO-COMPUTER-FIRMWARE-v1
ISSUE DATE  2026.08.05
STYLE       TERMINAL GRID
PARENT       docs/corporate/LOT-COSMO-COMPUTER-v1.md (Section 10, item 1)
SIBLING      docs/technical/LOT-COSMO-COMPUTER-SOFTWARE-v1.md (site-side half
              of the same LOT API connector – kept separate per brief point 11)
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00 SCOPE

This document specifies the firmware that runs ON the COSMO® Computer (ESP32-S3, per docs/corporate/LOT-COSMO-COMPUTER-v1.md Section 03). It does not specify server-side or lot-systems.com client-side behavior – that is the sibling SOFTWARE document. The two meet at exactly one boundary: the LOT API connector's wire format (Section 05 below), which both documents must agree on and neither owns unilaterally.

This is a PRE-HARDWARE specification. No firmware has been flashed to a physical unit as of this document's issue date. It exists to give the first physical prototype (parent doc Section 06, assembly step 4) a target to build against.

01 BOOT + PROVISIONING

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POWER ON (Qi charge detected OR battery has charge)
|
v
Read provisioning state from flash (NVS)
|
+-- UNPROVISIONED --> BLE ADVERTISE
|
|           Operator pairs the unit to their
|           lot-systems.com session via BLE from an
|           already-authenticated browser/app session
|           (parent doc Section 03: "BLE reserved for
|           v1.0 provisioning only"). Device receives:
|           - Wi-Fi credentials
|           - Per-device auth token (Section 05)
|           - Operator account binding
|           Writes to NVS, reboots into PROVISIONED path.
|
+-- PROVISIONED --> Connect Wi-Fi (2.4GHz) --> hold LOT API
                    connector session (Section 05) --> enter
                    MAIN LOOP (Section 02)
```

UNPAIR (operator-initiated, from lot-systems.com device settings, or long-press Copy button 10s): clears NVS provisioning state, returns to UNPROVISIONED. A COSMO® Computer with no bound operator account holds no session and displays nothing – mirrors LOT_ROBOTICS_COSMO.md's own hardware principle: "A COSMO® unit without a verified LOT profile does not activate."

02 MAIN LOOP

Runs continuously once provisioned. Four concurrent tasks on the ESP32-S3's dual cores, all published against the up/down channel defined in Section 05:

TASK A – DOWN-CHANNEL LISTENER

Holds the LOT API connector session (poll interval v1.0, per parent doc Section 05 transport note). On a pager-class notification payload: decode text (<=20 chars target), render to e-paper (Section 03), store the notification's ID as CURRENT_NOTIFICATION for the Copy button (Section 04) to reference.

TASK B – BUTTON WATCHER

GPIO interrupt on the Copy button. Debounce (hardware + 20ms firmware confirm). On confirmed press: emit a device event carrying CURRENT_NOTIFICATION's ID (Section 04/05).

TASK C – WEATHER POLL

BME280 read on a fixed low-frequency interval (v1.0 target: every

10-15 minutes – ambient data, not a live feed, matching parent doc Section 05's "passive, ambient, continuous-ish" characterization). Queues a telemetry event on the up-channel.

TASK D – CAMERA (GATED, OPERATOR-INITIATED ONLY)

No task runs this on a timer. Camera capture fires ONLY from an explicit operator action – a distinct long-press gesture reserved from the Copy button's short-press semantics (exact gesture TBD at first prototype; NOT a v1.0 motion trigger, NOT remote-initiated). This enforces the CAMERA GATE recorded in the parent document, Section 08 – loosening it is an explicit future decision, not a firmware default.

03 E-PAPER DRIVER

SPI display driver for the 1.54" e-paper module (parent doc Section 03). Partial-refresh mode for short text swaps (battery + speed); full refresh on boot and roughly every N partial refreshes to clear ghosting (exact N tuned at bring-up – e-paper panels vary by vendor batch).

Render contract: ONE short line of text per notification, no scrolling, no animation, no queue of unread messages shown at once. This is a firmware-level enforcement of the parent document's anti-feed principle (Section 02) – the display physically cannot become a notification list, because the driver only ever holds one string in its render buffer.

04 COPY BUTTON EVENT (BRIEF POINT 16)

ON confirmed press (Section 02, Task B):

IF CURRENT_NOTIFICATION is set:

build device_notification_copy event:

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{ device_id, notification_id: CURRENT_NOTIFICATION,
  copied_text: <the currently rendered string>,
  ts: <device RTC, corrected against last known-good
    server time> }
```

enqueue on up-channel (Section 05)

flash a brief e-paper acknowledgment (e.g. a corner glyph, not a new message – screen contract in Section 03 still holds)

ELSE:

no-op – nothing to copy if no notification is currently displayed. No error state shown; silence is correct here.

This firmware-side event is the device half of the signal path fully diagrammed in docs/corporate/LOT-COSMO-COMPUTER-v1.md Section 04. The server-side handler (new Logs.tsx military handler, working name COPY:) is sibling-document, not firmware, scope.

05 LOT API CONNECTOR – DEVICE-SIDE CLIENT

Firmware-side half only. Wire format, auth, and server routes are defined jointly with the sibling SOFTWARE document – this section states what the DEVICE must implement, not the full protocol.

- | | |
|--------------|---|
| SESSION | Per-device auth token issued at provisioning (Section 01), held in NVS, sent with every up/down-channel request. |
| DOWN-CHANNEL | HTTPS poll (v1.0) at a bounded interval (target: 15-30s when idle, tightened opportunistically after any recent activity – exact backoff curve tuned at bring-up). Payload: one pager-class notification (text, id) or empty. v1.1 upgrade to a persistent push channel is named in the parent doc roadmap (Section 07) and is NOT v1.0 firmware scope. |
| UP-CHANNEL | HTTPS POST, three event classes: <ol style="list-style-type: none">1. device_notification_copy (Section 04)2. weather_telemetry (Section 02, Task C)3. device_photo_capture (Section 02, Task D – gated, operator-initiated only) Queued locally if Wi-Fi is briefly unavailable; dropped (not indefinitely retried) past a bounded queue depth – a desk object with a battery buffer (parent doc Section 03) is not a store-and-forward log shipper. |
| FAILURE MODE | Loss of Wi-Fi or auth: e-paper holds its last rendered string (no error message shown – see Section 03 render contract) until connectivity restores. This is a |

deliberate quiet-failure design: an object whose whole purpose is to NOT compete for attention should not start competing for attention specifically to report that it is broken.

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06 WHAT THIS DOCUMENT DOES NOT SPECIFY

Exact GPIO pin assignments, clock configuration, RTOS task priorities, and power-management sleep-state tuning are bring-up work against a physical prototype board, not corpus-level planning. This document fixes the ARCHITECTURE (tasks, event classes, render contract, gates) that bring-up work must not violate – it is not itself a build-ready firmware image spec.

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